Pizza Sales Performance Analysis

Business Requirements Document (BRD)

Document Type	Business Requirements Document (BRD)
Project Name	Pizza Sales Performance Analysis
Prepared By	Project Analyst / Author
Version	1.0
Status	Final
Data Period Covered	January 2015 – December 2015
Primary Tools	Microsoft SQL Server (T-SQL), Microsoft Power BI

1. Introduction & Purpose

This document defines the business requirements for the Pizza Sales Performance Analysis project. The project analyzes one year of pizza-store transaction data to produce a set of standardized key performance indicators (KPIs) and an interactive Power BI dashboard that management can use to monitor sales performance, identify trends, and make informed decisions about menu, staffing, and promotions.

The requirements captured here were used to guide the SQL data preparation and the design of the Power BI report.

2. Business Background & Problem Statement

Prior to this project, pizza sales insight was fragmented across spreadsheets with no consistent definition of core metrics, making it difficult for leadership to track performance over time or compare it across categories, sizes, and menu items. The business requested a single, reliable source of truth for sales performance, built from the raw order data and presented through a dashboard that could be filtered and explored interactively.

Business objectives

- Establish a consistent, auditable definition for each core sales metric.
- Reveal daily, hourly, and monthly demand patterns to support staffing and inventory planning.
- Identify which pizza categories, sizes, and menu items drive — or drag down — performance.
- Provide a self-service dashboard so stakeholders can explore the data without writing SQL.

3. Scope

Area	Description
In Scope	Historical order-level pizza sales data for one calendar year; KPI calculation via T-SQL; Power BI data model, DAX measures, and interactive report; category, size, and time-based trend analysis; best/worst seller ranking.
Out of Scope	Real-time / streaming data feeds; customer-level or loyalty data; inventory and cost/margin analysis; predictive forecasting; integration with POS or ERP systems.

4. Stakeholders

Stakeholder	Interest / Use of the Report
Business Owner / Leadership	Sponsor of the analysis; primary consumer of the KPI summary and trend insights.
Store Operations Manager	Uses daily/hourly trend and busiest-day insights for staffing decisions.
Menu / Category Manager	Uses category, size, and best/worst seller insights for menu decisions.
Data Analyst (Project Author)	Designs the SQL logic, data model, and Power BI dashboard.

5. Data Source

A single flat-file (CSV) export of order-level pizza transactions was imported into a SQL Server database (**Pizza_Store**) as the table **pizza_sales**.

Field	Description
order_id	Unique identifier for a customer order (one order may contain multiple pizzas)
order_date	Calendar date the order was placed
order_time	Time of day the order was placed
pizza_name	Name of the pizza ordered
pizza_category	Menu category (e.g., Classic, Veggie, Chicken, Supreme)
pizza_size	Size of the pizza (Small, Medium, Large, etc.)
quantity	Number of units of that pizza in the order line
unit_price	Price per pizza unit
total_price	Extended price for the order line (quantity × unit_price)

6. Functional Requirements — KPIs

The business requested five core KPIs, each calculated directly from the pizza_sales table using T-SQL aggregate functions.

KPI	Definition	Calculation Logic	Result
Total Revenue	Sum of the total price of all pizza orders	SUM(quantity × unit_price)	$817.86K
Average Order Value (AOV)	Average amount spent per order	Total Revenue ÷ COUNT(DISTINCT order_id)	$38.31
Total Pizzas Sold	Sum of the quantities of all pizzas sold	SUM(quantity)	49,574
Total Orders	Total number of distinct orders placed	COUNT(DISTINCT order_id)	21,350
Average Pizzas per Order	Average number of pizzas sold per order	Total Pizzas Sold ÷ Total Orders	2.32

7. Functional Requirements — Visualizations

Seven visualizations were required to help the business understand trends and performance across time, category, size, and individual menu items.

Requirement	Chart Type	Business Question Answered
Daily Trend for Total Orders	Bar chart	Identify patterns or fluctuations in order volume by day of week
Hourly Trend for Total Orders	Line chart	Identify peak hours / periods of high order activity
Monthly Trend for Total Orders	Line chart	Show how order volume shifts across the year
% of Sales by Pizza Category	Pie chart	Show the distribution of sales across pizza categories
% of Sales by Pizza Size	Donut chart	Understand customer preference for pizza size
Total Pizzas Sold by Category	Funnel chart	Compare sales volume performance across categories
Top 5 Best Sellers	Bar chart	Highlight top 5 pizzas by revenue, quantity, and total orders
Bottom 5 Worst Sellers	Bar chart	Highlight bottom 5 pizzas by revenue, quantity, and total orders

8. Technical & Non-Functional Requirements

- **Data accuracy:** All KPI values must reconcile exactly with SQL aggregate query results.
- **Interactivity:** The dashboard must support filtering by pizza category and by date range.
- **Usability:** Navigation between the Home and Best/Worst Seller pages must be a single click.
- **Performance:** Report visuals must load and respond to filter changes without noticeable lag for a dataset of this size (~21K orders / ~49K line items).
- **Maintainability:** SQL queries should be modular and documented so metrics can be re-derived or audited independently of the Power BI layer.

9. Assumptions & Constraints

Type	Detail
Assumption	The supplied CSV/order data is complete and accurate for the full Jan–Dec period.

Type	Detail
Assumption	"Revenue" is defined as quantity × unit price, with no tax, discount, or refund adjustments in the source data.
Constraint	Analysis is retrospective (historical) only — no live or real-time data connection.
Constraint	Report is built and delivered as a single Power BI file (.pbix) with a single fact table.

10. Deliverables

- T-SQL query script covering all five KPIs and all seven chart datasets.
- Power BI (.pbix) interactive dashboard with Home and Best/Worst Seller pages.
- This Business Requirements Document (BRD).
- Summary of key business insights derived from the analysis.

11. Acceptance Criteria

- Every KPI on the dashboard matches the corresponding SQL query result.
- All seven required chart types are present and correctly filtered by the category/date slicers.
- The Best/Worst Seller page correctly ranks pizzas by revenue, quantity, and order count.
- The report is reviewed and approved by the business owner as ready for interview/portfolio use.

End of document.